Name ____ Date ___________ Period _____

Domain and Range: All Three Notations - Practice

Algebra II Honors | Unit 1 | Lesson 1 Practice | TEK A2.7.I

Objective: I can find domain and range from pairs, tables, and graphs; determine a domain algebraically from a rule; and use union notation for a split domain.

Section 1 — Recall

1. { (-7, 1), (-2, -4), (0, 1), (5, 9) } - give D and R in set notation, and name which value repeats and why the range still lists it once.

D: { ______________________________ } R: { __________________________ }

Repeats: ____

2. Domain of $h(x) = \sqrt{x - 6}$ - inequality and interval notation.

Inequality: _______________ Interval: _______________

3. Domain of $k(x) = 1/(x + 4)$ - inequality and interval (union) notation.

Inequality: _______________ Interval: ________________________________

4. Domain of $j(x) = \sqrt{10 - x}$ - inequality and interval notation.

Inequality: _______________ Interval: _________________

Section 2 — Apply

5. Domain of $m(x) = 1/\sqrt{x - 2}$ - inequality and interval notation, plus one sentence on why the boundary point itself must be excluded.

Inequality: _____________ Interval: _______________

Why the boundary: ____

6. Domain of $p(x) = (x - 1)/(x^2 - 16)$ - show the factoring, then give the domain in interval AND set notation.

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Factors as ________________________ , so x ___________

Interval: ____

Set notation: ______________________________________

7. Domain of $r(x) = 1/(x^2 - 5x + 6)$ - factor to find both excluded values, then give the domain in interval notation.

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Factors as ________________________ , so x ___________

Interval: ____

8. The domain $(-\infty, -1) \cup (3, \infty)$ excludes every value from -1 to 3. Explain why this cannot be written as a single inequality chain (like $a \le x \le b$), then write the two inequalities that together describe it.

Why not one chain: ____

Two inequalities: __________________________

Section 3 — Challenge

9. Write a function whose domain is exactly $(-\infty, 5) \cup (5, \infty)$.

Sample: ___________________________ (any function excluding only x = 5 works)

10. Write a function whose domain is exactly $(-\infty, -2) \cup (-2, 1) \cup (1, \infty)$.

Sample: ___________________________________ (any function excluding only x = -2 and x = 1 works)

11. Domain of $q(x) = \sqrt{(x + 3)}/(x - 1)$ - combine the radicand restriction AND the denominator restriction. Give the domain in interval notation and justify both restrictions in one sentence each.

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Radicand: ____ Denominator: ____________

Domain: _______________________________

Section 4 — Explain It

12. A student claims every function of the form 1/(quadratic expression) will have a domain written with exactly one union symbol, since there is exactly one denominator. Is the student correct? Explain, and describe what happens to the domain notation if the quadratic in the denominator has a repeated root, or no real roots at all.

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